Hamdan bin Rashid Al Maktoum Foundation for Medical and Educational Sciences (HF 2026)
1457-BRH Best Research in Healthcare

LQJJLnRB

Dravet Syndrome Modelling and mRNA Drug Design Reverses Molecular Phenotype in iPSC Derived Neurons

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Curriculum Vitae of First Author

PDF	Noor Kosaji _ CV.pdf (76 KiB download)

Curriculum Vitae of PI

DOCX	Mohammed Jashim Uddin CV.... (53 KiB download)

Additional Curriculum Vitae	Yes

Additional Curriculum Vitae #2

PDF	Shahed-CV.pdf (222 KiB download)

No Objection Letter

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First Page

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Research Paper

DOCX	Dravet Syndrome Modelling... (20.1 MiB download)

Supplementary Materials

DOCX	Supplementary.docx (9.9 MiB download)

Has the submitted research received any awards?	No

Layperson's Summary

> Dravet syndrome is a severe form of epilepsy that begins in infancy and is caused by changes in a gene called SCN1A. Children with this condition often experience frequent, hard-to-control seizures, developmental delays, and a high risk of early death. Currently, treatments are limited and mainly focus on reducing symptoms.

In this study, researchers analyzed the DNA of thousands of children with neurodevelopmental disorders to find changes that might explain their conditions. One important gene they focused on was SCN1A, which is known to be involved in epilepsy. To understand how changes in this gene affect brain cells, the team used advanced stem cell techniques to create human neurons with the SCN1A mutation in the lab.

These lab-grown neurons showed abnormal electrical activity and molecular changes similar to those seen in patients with Dravet syndrome. Using cutting-edge technologies, the researchers studied how the mutation affected genes, proteins, and cell metabolism.

To explore a potential treatment, they tested a new strategy: turning off the faulty gene using small molecules called antisense oligonucleotides (ASOs) and reintroducing a healthy version of the gene using a custom-built vector. This approach showed early promise in correcting the cellular problems caused by the mutation.

While still in progress, this work brings scientists closer to personalized treatments that directly target the cause of Dravet syndrome and may offer hope to families affected by this devastating condition.

Relevance - 10%

/ 5.00

5 being the highest and 1 as the lowest. * The research topic is highly relevant to the respective field. * The research topic addresses a significant gap or problem in healthcare within its respective field. * The research demonstrates potential for practical application or impact on current issues.

Study design - 25%

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5 being the highest and 1 as the lowest. * The aim of the research is clear. * The methodology used is appropriate to the research question(s) and explained well. * The study design minimizes bias and ensures the validity of the results. * Data collection methods (if applicable) are clearly defined. * Statistical and analytical methods are appropriate and well executed.

Results and findings - 25%

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5 being the highest and 1 as the lowest. * The results address the research question. * Findings are well presented, and the implications are discussed. * The results contribute to new knowledge to the field. * Findings are robust and reproducible.

Impact potential - 20%

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Presentation and clarity - 20%

/ 5.00

5 being the highest and 1 as the lowest. * The paper is well organized, clear, and concise. * The visual aids enhance the understanding of the research. * Figures and tables are clear and effective at presenting data. * The sources are relevant and sufficiently recent. * The writing style is professional.

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